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# SM Homework #7

prob 1.)

a.)

- single particle state can be occupied by at most M-particles.
- $M \geq 1$, $M = 1$ and $M = \infty$. (recover useful-fermions and Bosons)
- noninteracting, states $i$, have energy $\varepsilon_i$, equilib at temp $T$, and $\mu$.
- Use grand-canonical ensemble:

(chandler)

$$\Theta = \prod_j \left\{ \sum_{n_j=0}^{M} e^{-\beta(\varepsilon_j - \mu) n_j} \right\}$$

- $j \rightarrow$ quantum state
- $n_j \rightarrow$ occupation number
- $0 \leq n_j \leq M$
- $j \rightarrow$ states
- $\sum_{n=0}^{k} r^n = \frac{r^{k+1} - 1}{r - 1}$

$$\Theta = \prod_j \left( \frac{e^{-\beta(\varepsilon_j - \mu)(M+1)} - 1}{e^{-\beta(\varepsilon_j - \mu)} - 1} \right)$$

# General case:

b.)

- $\langle n_j \rangle = \dfrac{\partial \ln\Theta}{\partial(-\beta\varepsilon_j)} = -\dfrac{\partial \ln\Theta}{\partial(\beta\varepsilon_j)}$

- $0 \le j \le \infty$ particle states.

$$\ln\Theta = -\sum_j \ln\left[1 - e^{\beta(\mu-\varepsilon_j)}\right]$$

(Bosons)

$$\ln\Theta = \sum_j \left( \ln\left(e^{-\beta(\varepsilon_j-\mu)(M+1)} - 1\right) - \ln\left(e^{-\beta(\varepsilon_j-\mu)} - 1\right)\right)$$

Fugacity: $Z = e^{\beta\mu}$

$$= \sum_j \left( \ln\left(e^{-\beta\varepsilon_j(M+1)} e^{\beta\mu(M+1)} - 1\right) - \ln\left(e^{-\beta\varepsilon_j} e^{\beta\mu} - 1\right)\right)$$

$$= \sum_j \left( \ln\left(Z^{(M+1)} e^{-\beta\varepsilon_j(M+1)} - 1\right) - \ln\left(Ze^{-\beta\varepsilon_j} - 1\right)\right)$$

- $\ln\Theta = \ln\left(Z^{(M+1)} e^{-\beta\varepsilon_j(M+1)} - 1\right) - \ln\left(Ze^{-\beta\varepsilon_j} - 1\right)$ — $j^{th}$ state:

- $\ln\Theta \simeq \ln\left(z^{(M+1)} e^{-\beta\varepsilon_j(M+1)} - 1\right) - \ln\left(ze^{-\beta\varepsilon_j} - 1\right)$

- $\frac{\partial \ln\Theta}{\partial(\beta\varepsilon_j)} = \frac{1}{\left(z^{(M+1)} e^{-\beta\varepsilon_j(M+1)} - 1\right)} \cdot \left(-(M+1) z^{(M+1)} e^{-\beta\varepsilon_j(M+1)}\right)$

$$- \frac{1}{\left(ze^{-\beta\varepsilon_j} - 1\right)} \cdot \left(-ze^{-\beta\varepsilon_j}\right)$$

- $\frac{\partial \ln\Theta}{\partial(\beta\varepsilon_j)} = \frac{-(M+1) z^{(M+1)} e^{-\beta\varepsilon_j(M+1)}}{\left(ze^{-\beta\varepsilon_j(M+1)} - 1\right)} + \frac{ze^{-\beta\varepsilon_j}}{\left(ze^{-\beta\varepsilon_j} - 1\right)}$

- $\boxed{\langle n_j \rangle = \frac{(M+1) z^{(M+1)} e^{-\beta\varepsilon_j(M+1)}}{\left(z^{(M+1)} e^{-\beta\varepsilon_j(M+1)} - 1\right)} - \frac{ze^{-\beta\varepsilon_j}}{\left(ze^{-\beta\varepsilon_j} - 1\right)}}$

- $\beta = \frac{1}{k_B T}$ High $T$, $\beta \to 0$ :

  - $e^{-\beta\varepsilon_j} \simeq 1 \quad (\beta \to 0)$
  - $e^{-\beta\varepsilon_j(M+1)} \simeq 1 \quad (\beta \to 0)$
  - $e^{\beta' M} \simeq 1 \quad (\beta \to 0)$

# Power - Series expansions:

- $\sum_{n=0}^{\infty} \frac{f^n(0)}{n!} x^n$ (Mcclaurin series)

- $\sum_{n=0}^{\infty} \frac{f^n(X_a)}{n!} (x - X_a)^n$ (Taylor Series)

- $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$

- $T \to \infty, \beta \to 0$ (1-3 terms in expansions)

- $e^{ax} = \sum_{k=0}^{\infty} \frac{(ax)^k}{k!}$

- $e^{\beta M} \simeq 1 + \beta M \propto Z$
- $e^{-\beta \varepsilon_j} \simeq 1 - \beta \varepsilon_j$
- $e^{-\beta \varepsilon_j (M+1)} \simeq 1 - \beta \varepsilon_j (M+1)$

$$\langle n_j \rangle \cong \frac{(M+1)\left(1+\beta M\right)^{(M+1)}\left(1-\beta\varepsilon_j(M+1)\right)}{\left(\left(1+\beta M\right)^{(M+1)}\left(1-\beta\varepsilon_j(M+1)\right)-1\right)} - \frac{\left(1+\beta M\right)\left(1-\beta\varepsilon_j\right)}{\left(\left(1+\beta M\right)\left(1-\beta\varepsilon_j\right)-1\right)}$$

* Low temperature :

• $T \to 0$, $\beta \to \infty$ * •

$$\bullet \langle n_j \rangle = \frac{(M+1)Z\, e^{(M+1)-\beta\varepsilon_j(M+1)}}{\left(Z^{(M+1)}e^{-\beta\varepsilon_j(M+1)}-1\right)} - \frac{Ze^{-\beta\varepsilon_j}}{\left(Ze^{-\beta\varepsilon_j}-1\right)}$$

$$= \frac{(M+1)\,e^{-\beta(\varepsilon_j-M)(M+1)}}{\left(e^{-\beta(\varepsilon_j-M)(M+1)}-1\right)} - \frac{e^{-\beta(\varepsilon_j-M)}}{\left(e^{-\beta(\varepsilon_j-M)}-1\right)}$$

$$= \frac{(M+1)}{\left(1-e^{\beta(\varepsilon_j-M)(M+1)}\right)} - \frac{1}{\left(1-e^{\beta(\varepsilon_j-M)}\right)}$$

- very large approximation for exponential!

- (Large n.) $\left(1+\frac{x}{n}\right)^n \approx e^x$ (Desmos to check)

- $$\langle n_j \rangle = \frac{(M+1)}{\left(1-\left(\frac{1+\beta(\xi_j-\mu)}{n}(M+1)\right)^n\right)} - \frac{1}{\left(1-\left(1+\frac{\beta(\xi_j-\mu)}{n}(M+1)\right)^n\right)}$$

c.) $$\langle n_j \rangle = \frac{(M+1)}{\left(1-\exp\left(\beta(\xi_j-\mu)(M+1)\right)\right)} - \frac{1}{\left(1-\exp\left(\beta(\xi_j-\mu)(M+1)\right)\right)}$$

$T \to 0$, $\beta \to \infty$

- $\xi_j > M$ : $\langle n_j \rangle = 0$

- $\xi_j < M$, $(M+1) - 1 = M$ ; $\langle n_j \rangle = M$

$$T = 0K \quad \langle n_j \rangle = \begin{cases} M & \xi_j < M \\ 0 & \xi_j > M \end{cases}$$

Prob 2.)

$\varepsilon_i \rightarrow$ Internal energy.
$\varepsilon_k \rightarrow$ kinetic energy.

a) $\varepsilon = \varepsilon_k + \varepsilon_i$

- Ideal Bose-Gas in 3-D $N = \int_0^\infty g(\varepsilon) \frac{1}{\exp(\beta(\varepsilon-\mu)) - 1} d\varepsilon$

- $g(\varepsilon) = \frac{V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \varepsilon^{1/2}$, $\varepsilon \geq 0$

$$N_e = \int_0^\infty \frac{V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \frac{\varepsilon^{1/2} d\varepsilon}{\left(\exp(\beta(\varepsilon_k + \varepsilon_i - \mu))\right) - 1}$$

- $z = e^{\beta\mu}$

- $\frac{1}{z} = e^{-\beta\mu} = z^{-1}$

- Gas is made up of structe and structellts particles.

$$N_e = \frac{V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \int_0^\infty \frac{\varepsilon^{1/2} d\varepsilon}{\left(z^{-1} e^{\beta\varepsilon} - 1\right)} + \frac{V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \int_0^\infty \frac{\varepsilon^{1/2} d\varepsilon}{\left(z^{-1} e^{\beta\varepsilon_k} e^{\beta\varepsilon_i}\right) - 1}$$

$$I_1 = \int_0^\infty \frac{\varepsilon^{1/2}}{\left(z^{-1} e^{\beta\varepsilon} - 1\right)}$$

- $x = \beta\varepsilon$
- $dx = \beta d\varepsilon$
- $\frac{x}{\beta} = \varepsilon$

$$I_1 = \left(\frac{1}{\beta}\right)^{1/2} \frac{1}{\beta} \int_0^\infty \frac{x^{1/2} dx}{\left(z^{-1} e^{x} - 1\right)}$$

$$I_1 = \frac{1}{\beta^{3/2}} \int_0^\infty \frac{x^{1/2}}{\left(z^{-1} e^{x} - 1\right)} = \frac{1}{\beta^{3/2}} g_{\frac{3}{2}}(z)\,\Gamma\left(\frac{3}{2}\right)$$

$$Z_\alpha = \exp\left(\beta(-\mu+\varepsilon_1)\right)$$

$$\frac{1}{Z_\alpha} = \exp\left(\beta(\varepsilon-\mu)\right)$$

- $$I_2 = \int_0^\infty \frac{\varepsilon^{1/2}\, d\varepsilon}{\left(\exp\left(\beta(-\mu+\varepsilon_1)\right)\exp(\beta\varepsilon) - 1\right)}$$

- $$\int_0^\infty \frac{\varepsilon^{1/2}\, d\varepsilon}{Z_\alpha^{-1}\exp(\beta\varepsilon) - 1} = I_2$$

$\Gamma(3/2) = \frac{\sqrt{\pi}}{2}$ · $x = \beta\varepsilon$ · $dx = \beta d\varepsilon$

$$\frac{1}{\beta^{3/2}}\int_0^\infty \frac{x^{1/2}\, dx}{Z_\alpha^{-1}\exp(x) - 1} = I_2$$

- $$\frac{1}{\beta^{3/2}} g_{3/2}(z_\alpha)\Gamma\left(\frac{3}{2}\right) = I_2$$

- $$N_c = \frac{V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\frac{1}{\beta^{3/2}} g_{3/2}(z)\Gamma\left(\frac{3}{2}\right) + \frac{V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\frac{1}{\beta^{3/2}} g_{3/2}(z_\alpha)\Gamma\left(\frac{3}{2}\right)$$

$$= \frac{V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\left(k_B T\right)^{3/2}\frac{\pi^{1/2}}{2} g_{3/2}(z)$$

$$+ \frac{V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\left(k_0 T\right)^{3/2}\frac{\pi^{1/2}}{2} g_{3/2}(z_\alpha)$$

$$\frac{V 2^{3/2} m^{3/2} k_B^{3/2} T^{3/2} \pi^{1/2} g_{3/2}(z)}{2^2 \pi^2 \hbar^3}$$

$$\frac{V m^{3/2} k_B^{3/2} T^{3/2} g_{3/2}(z)}{2^{-3/2}\pi^{-3/2}\hbar^3}$$

$\uparrow \lambda_c^3$

- $\frac{1}{\lambda_T^3} = \left(\frac{m^{3/2} k_B^{3/2} T^{3/2}}{2^{-3/2} \pi^{-3/2} \hbar^3}\right)$ $\quad y = Ze^{-\varepsilon_1/k_B T_c}$

- $N_e = \frac{V}{\lambda_T^3} g_{3/2}(Z) + \frac{V}{\lambda_T^3} g_{3/2}(Z_\alpha)$

  - $T = T_c$, $Z \to 1$, and $N \to N_e$

- $N = \frac{V}{\lambda_c^3} g_{3/2}(1) + \frac{V}{\lambda_c^3} g_{3/2}(y)$

  ) $\lambda_c \to$ critical temperature

) $\varepsilon_1 \gg k_B T_c$, $y \ll 1$ and

small approx $g_{\frac{3}{2}}(y) \approx y$

) $N = \frac{V}{\lambda_c^3} \zeta\left(\frac{3}{2}\right) + \frac{V}{\lambda_c^3} y$ ( structed particle )

) $N = \frac{V}{\lambda_0^3} \zeta\left(\frac{3}{2}\right)$ ( structureless particle )

get rid of y-term.

- $\frac{1}{\lambda_0^3} \zeta\left(\frac{3}{2}\right) = \frac{V}{\lambda_c^3}\left(\zeta\left(\frac{3}{2}\right) + y\right)$

- $\left(\frac{\lambda_0}{\lambda_c}\right)^3 = \frac{\zeta\left(\frac{3}{2}\right)}{\left(\zeta\left(\frac{3}{2}\right) + y\right)}$ $\quad \frac{\lambda_0}{\lambda_c} = \frac{h}{(2\pi m k_B)^{1/2} T_0^{1/2}}$

$$T_c = T_0 \left[\frac{\zeta(3/2)}{\zeta(3/2) + y}\right]^{2/3} \qquad \frac{\lambda_0}{\lambda} = \left(\frac{T_c}{T_0}\right)^{1/2}$$

$\times \; (2\pi m k_B)^{1/2} T_c^{1/2} / h$

Prob 3.) $n_x, n_y, n_z = 0, 1, 2, \ldots$

a.) • anisotropic oscillator:

• For a harmonic trap the geometry for 2-D is triangle and 3-D is triangular prism.

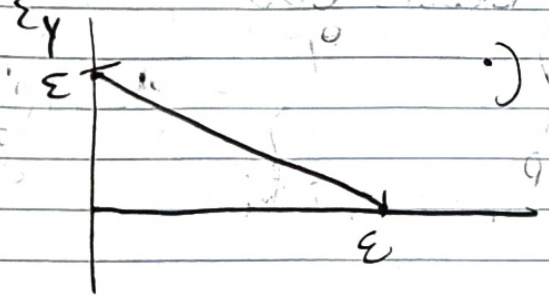
• only each state can have a highest energy of $\mathcal{E}$.

• (Reason is because of total collection of bosons in ground state and I nearly disappear out when I increase)

• lattice is triangle in this case.

∴ Triangle because of harmonic-oscillator

2-D:




• $A = \int_0^{\mathcal{E}} \mathcal{E}_x \, d\mathcal{E}_x = \frac{\mathcal{E}^2}{2}$

• $\mathcal{E}_y = \frac{\omega_y \left(n_y + \frac{1}{2}\right)}{\omega_x \left(n_x + \frac{1}{2}\right)}$

• $\text{Area} = \frac{1}{2}\mathcal{E}^2$ $\quad V = \frac{1}{2}\int_0^{\mathcal{E}} \mathcal{E}_z^2 \, d\mathcal{E}_z = \frac{1}{2}\frac{1}{3}\mathcal{E}^3$

$$V = \frac{1}{6}\mathcal{E}^3$$

1-state: $\hbar^3 \omega_x \omega_y \omega_z$, $\hbar^3 \omega_0^3 = V_{1\text{-state}}$

$$N = \frac{\text{Volume of total of states}}{\text{Volume of 1-state}}$$

$$= \#\ \text{energy states}$$

$$N = \frac{1}{6} \frac{\varepsilon^3}{(\hbar\omega)^3}$$

b.) $$\frac{dN}{d\varepsilon} = \frac{3}{6} \frac{\varepsilon^2}{(\hbar\omega_0)^3} = \boxed{\frac{1}{2} \frac{\varepsilon^2}{(\hbar\omega_0)^3} = g(\omega)}$$

·) BEC occurs if $z \to 1$ — $N_e$ is finite and bounded

$$z^{-1} = \exp(-\beta\mu)$$

$$N_e = \int_0^\infty g(\varepsilon) \langle n_{FD} \rangle \, d\varepsilon$$

$$N_e = \frac{1}{2(\hbar\omega_0)^3} \int_0^\infty \frac{\varepsilon^2}{\exp(\beta(\varepsilon - \mu)) - 1} \, d\varepsilon$$

$$N_e = \frac{1}{2(\hbar\omega_0)^3} \int_0^\infty \frac{\varepsilon^2}{z^{-1}\exp(\beta\varepsilon) - 1}$$

- $x = \beta\varepsilon$, $\frac{x}{\beta} = \varepsilon$
- $dx = \beta \, d\varepsilon$

$$N_e \cong \frac{1}{2(\hbar\omega_0)^3 \beta^3} \int_0^\infty \frac{x^2}{z^{-1}\exp(x) - 1}$$

- $I = \int_0^\infty \frac{x^2\,dx}{z^{-1}e^x - 1}$

- $V - 1 = 2^0$

$V = 3$

$\Gamma(3)\,g_3(z) = I = \int 2\,g_3(z) \rightarrow 2\underbrace{\zeta(3)}_{1.202} = 2.404$

$z \to 1$

BEC occurs because $N_e \approx 2.404$ is finite and bounded

$N \approx 2.404 \approx$

$N = 2\zeta(3)\frac{1}{\pi(\hbar\omega_0\beta)^3}$

$\left(\frac{N}{\zeta(3)}\right)^{1/3} \frac{\hbar\omega_0}{k_B} = T_c$

# Prob 4.)

- $q = q_{trans} \cdot q_{Rot} \cdot q_{vibr}$
- $\boxed{q_{trans}(T,V) = V\left(\frac{2\pi m}{\beta h^2}\right)^{3/2}}$
- $q_{rot}(T) = \sum_{J=0}^{\infty} (2J+1)\exp\left(-J(J+1)\beta\hbar^2/2I_0\right)$
  - $\varepsilon_r = \frac{\hbar^2}{2I_0}$, $T \gg \frac{\varepsilon_r}{k_B}$ then transform this into an integral
- $q_{rot}(T) \cong \int_0^{\infty} (2J+1)\exp\left(-J\beta(J+1)\varepsilon_r\right) dJ$ .
  - $\int_0^{\infty} (2x+1)\exp\left(-a(x+1)\right) dx = I$
  - $x = -J\beta\varepsilon_r(J+1)$
  - Put in wolfram alpha
  - $dx = -\beta\varepsilon_r(2J+1)\,dJ$

$$q_{rot}(T) \approx \frac{-1}{\beta\varepsilon_r}\int \exp(x)\,dx$$

$$\approx \frac{-1}{\beta\varepsilon_r}\exp\left(-J\beta\varepsilon_r(J+1)\right)\Big|_0^{\infty}$$

$$\approx \frac{-1}{\beta\varepsilon_r}\left[0 - 1\right] = \frac{1}{\beta\varepsilon_r}$$

$$q_{rot}(T) \approx k_B T \frac{2I_0}{\hbar^2}$$

$$q_{vib} \approx \sum_{v=0}^{\infty} \exp\left(-\left(\frac{1}{2}+v\right)\beta\hbar\omega_0\right)$$

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

$$q_{vib} \approx \exp\left(-\frac{1}{2}\beta\hbar\omega_0\right) \underbrace{\sum_{v=0}^{\infty} \exp\left(-\beta v\hbar\omega_0\right)}$$

$$\frac{1}{1-\exp(-\beta\hbar\omega_0)}$$

$$q_{vib} \approx \frac{1}{\exp\left(\frac{\beta\hbar\omega_0}{2}\right)\left(1-\exp(-\beta\hbar\omega_0)\right)}$$

$$\approx \frac{1}{\left(\exp\left(\frac{\beta\hbar\omega_0}{2}\right) - \exp\left(-\frac{\beta\hbar\omega_0}{2}\right)\right)}$$

$$\approx \frac{1}{2\sinh\left(\frac{\beta\hbar\omega_0}{2}\right)}$$

$$q_{vib}(T) \approx \frac{1}{2\sinh\left(\frac{\beta\hbar\omega_0}{2}\right)}$$

$$Q = V\left(\frac{2\pi m}{\beta h^2}\right)^{3/2} k_B T \frac{2I_0}{\hbar^2} \frac{1}{2\sinh\left(\frac{\beta\hbar\omega_0}{2}\right)}$$

apply Indistinguishability:

$$Q = \frac{1}{N!} V^N \left(\frac{2\pi m}{\beta h^2}\right)^{3N/2} (k_B T)^{3N/2} \left(\frac{2I_0}{\hbar^2}\right)^N \left(\frac{1}{2\sinh\left(\frac{\beta\hbar\omega_0}{2}\right)}\right)^N$$

- $\langle E \rangle = -\frac{\partial \ln Q}{\partial \beta}$

- $Q = \frac{V^N \left(\frac{2\pi m}{\beta h^2}\right)^{\frac{3N}{2}} \left(\frac{2I_0}{\hbar^2}\right)^N \left(\frac{1}{2\sinh\left(\frac{\beta\hbar\omega_0}{2}\right)}\right)^N (k_B T)^N}{N!}$

- $\ln Q = \ln\left(V^N\right) + \ln\left(\left(\frac{2\pi m}{\beta h^2}\right)^{\frac{3N}{2}}\right) + \ln\left(\left(\frac{2I_0}{\hbar^2}\right)^N\right) + \ln\left(\left(\frac{1}{2\sinh\left(\frac{\beta\hbar\omega_0}{2}\right)}\right)^N\right) - \ln(N!)$

- $\ln Q = N\ln(V) + \frac{3N}{2}\ln(2\pi m) - \frac{3N}{2}\ln(\beta h^2) + N\ln(2I_0) - N\ln(\hbar^2) + N\ln\left(2\sinh\left(\frac{\beta\hbar\omega_0}{2}\right)\right) - \ln(N!) + N\ln(\beta)$

- $\sinh(x) = \frac{e^x - e^{-x}}{2}$

- $\frac{\partial}{\partial x}\left(\sinh(x)\right) = \frac{(2)\left(e^x + e^{-x}\right)}{4} = \frac{e^x + e^{-x}}{2} = \cosh(x)$

- $\frac{\partial \ln(Q)}{\partial \beta} = \frac{-3N}{2}\left(\frac{1}{\beta\hbar^2}\right)\cdot\hbar^2 - N\left(\frac{1}{2\sinh\left(\frac{\beta\hbar\omega_0}{2}\right)}\right)$

$T^{-1}$ $T^{-2}$

$\cdot 2\cosh\left(\frac{\beta\hbar\omega_0}{2}\right)\cdot\frac{\hbar\omega_0}{2}$

$$= \frac{-3N}{2\beta} - \frac{N\hbar\omega_0\cosh\left(\frac{\beta\hbar\omega_0}{2}\right)}{2\sinh\left(\frac{\beta\hbar\omega_0}{2}\right)} - \frac{N}{\beta}$$

$$= \frac{-3N}{2\beta} - \frac{N\hbar\omega_0}{2}\coth\left(\frac{\beta\hbar\omega_0}{2}\right) - \frac{N}{\beta}$$

$$= \frac{-3N}{2}k_BT - \frac{N\hbar\omega_0}{2}\coth\left(\frac{\beta\hbar\omega_0}{2}\right) - Nk_BT$$

$$\boxed{\langle E\rangle = \frac{3}{2}Nk_BT + \frac{N\hbar\omega_0}{2}\coth\left(\frac{\hbar\omega_0}{2k_BT}\right) + Nk_BT}$$

$\frac{\cosh(x)}{\sinh(x)} = \frac{\sinh^2(x) - \cosh^2(x)}{\sinh^2}$

- $C_V = \frac{\partial\langle E\rangle}{\partial T}$

- $C_V = \frac{3}{2}Nk_B + Nk_B + \frac{N\hbar\omega_0}{2}\left(\frac{\sinh^2\left(\frac{\hbar\omega_0}{2k_BT}\right) - \cosh^2\left(\frac{\hbar\omega_0}{2k_BT}\right)}{\sinh^2\left(\frac{\hbar\omega_0}{2k_BT}\right)}\right)\cdot\frac{-\hbar\omega_0}{T^2k_B}$